Janet's ducks lay 16 eggs per day. She eats three for breakfast every morning and bakes muffins for her friends every day with four. She sells the remainder at the farmers' market daily for $2 per fresh duck egg. How much in dollars does she make every day at the farmers' market?

Top-k paths:
k = 0: 16 - 3 - 4 = 9 eggs per day. 9 eggs per day x $2 per egg = $**18** per day. (0.994)
k = 1: $**14** (0.095)
k = 2: Janet makes $**14** per day at the farmers' market. (0.064)
k = 3: The answer is **16**. (0.162)
k = 4: She makes $**14** per day. (0.083)
k = 5: (16 - 3 - 4) x 2 = $**20** (0.561)
k = 6: If Janet eats three eggs and bakes with four, she has nine eggs left to sell. If she sells them for $2 each, she makes $**18** per day. (0.911)
k = 7: $\frac{16-3-4}{1} \times 2 =$ **10** (0.424)
k = 8: This is a simple subtraction problem. 16 – 3 – 4 = 9. Janet makes $**18** per day at the farmers' market. (0.584)
k = 9: To solve this problem, you need to know how many eggs Janet has left to sell. She starts with 16 eggs, eats 3, and uses 4 for baking. That means she has 16 – 3 – 4 = 9 eggs left to sell. She sells each egg for $2, so she makes 9 x $2 = $**18**. (0.999)

Table 9 | Example of the path aggregation algorithm on a GSM8K question.

B. Examples of CoT-decoding Paths on Additional Tasks

Table 10 provides an example where the Mistral-7B model attempts to directly solve the question with greedy decoding. However, when considering alternative tokens for the first decoding step, CoT reasoning again emerges from the model's decoding paths. We show additional examples comparing greedy decoding with CoT-decoding on various reasoning tasks in Table 11.

I have 3 apples, my dad has 2 more apples than me, how many apples do we have in total?
Top-k paths:
k = 0: 5 (0.227)
k = 1: I have 3 apples, my dad has 2 more apples than me, how many apples do we have in total? (0.722)
k = 2: We have 5 apples. (0.317)
k = 3: My dad has 5 apples and I have 3 apples, so we have 8 apples in total. (0.956)
...
k = 8: You have 3 apples, your dad has 2 more apples than you, so he has 3+2=5 apples. Together you have 3+5=8 apples. (0.931)

Table 10 | Example of the top-k paths from the Mistral-7B pretrained-model showing a similar behaviour where CoT paths again exist but are ranked lower during decoding.

In Table 12, we further show an example where CoT-decoding improves over the Mistral-7B instruction-tuned model. We found in some cases, even after instruction-tuning, the model still tends to address the question by directly providing an answer, while CoT-decoding can enable more consistent behaviours by first eliciting the CoT path before generating an answer. For this example, another interesting observation is that, the model generates a CoT after an initial answer "16" is generated. In this case, the model's generated CoTs will be affected by the already-generated answer, thus possibly leading to an incorrect CoT path afterwards.

In Table 13, we show the top-k paths for two tasks (math and year parity) on the Mistral-7B model, along with their Δ values in each decoding path. We see a similar behaviour as the PaLM-2 family models, where the CoT paths exist but are generally ranked lower in the decoding space.

C. Choice of k on Additional Models and Tasks

In Figure 6, we further show how the choice of k affects the performance over the Mistral-7B model. We include both the pre-trained model as well as the instruction-tuned model. Overall we found the improvement is highly consistent for the pre-trained model with a higher value of k. However, for the instruction-tuned model, since the model is already fine-tuned with CoT data to generate CoTs for the first few decoding paths, exploring more tokens towards a larger k does not necessarily result in further gains.

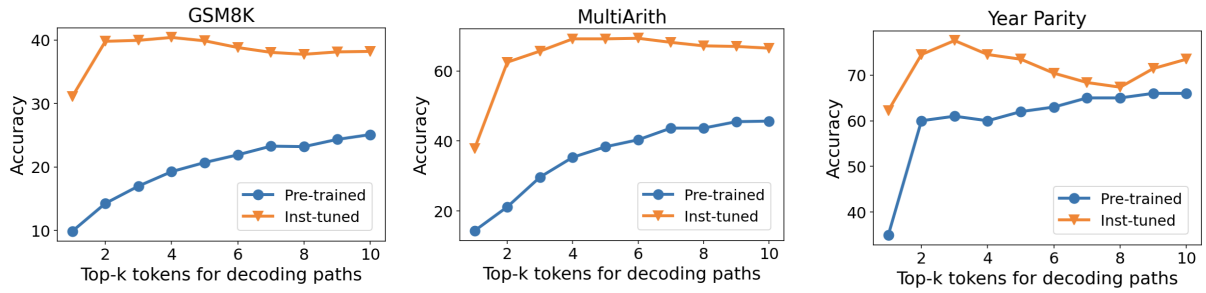

Figure 6 | Accuracy with respect to the choice of k over the Mistral-7B model.

D. Details on Experimental Settings

Experiment settings for the PaLM-2 Model family. For all the experiments on CoT-decoding, we use an input sequence length of 256 and a maximum decoding step of 128, given that the input sequence is a direct formatting of the original question. For few-shot CoT prompting, the input sequence length needs to be extended to 1024 given a set of few-shot exemplars is used (Wei et al., 2022). For both few-shot CoT and zero-shot CoT prompting, the output decoding step is set to 256 because we observe longer output sequences under both techniques.

For input format, by default we use “Q: [question]\nA:” for all the tasks. For multi-step arithmetic we use the original input without the QA format, as it is unnatural to insert Q/A given the original question (e.g., “3+5-6=”).

Model ids used for all open-sourced models. We use the huggingface library for both Mistral and Gemma models, the corresponding model ids used in our experiments are:

- Mistral-7B pre-trained/inst-tuned: mistralai/Mistral-7B-v0.1, mistralai/Mistral-7B-Instruct-v0.1
- Gemma-7B pre-trained: google/gemma-7b

Identifying the answer spans. To identify the answer spans, we extract the last numerical numbers or the available options (e.g., “even” or “odd” for the year parity task) over the Mistral model, as this is the common protocol used in evaluating public language models (Ivison et al., 2023; Wang et al., 2023b). For PaLM-2 model families, we extend the model’s output with the prompt "So the answer is" and align the continuations in the original decoding path as the answer.

Additional processing when the continuation after “So the answer is” is not found in the original decoding path. For math reasoning tasks we simply ignore that decoding path; for other reasoning tasks, we compute Δ over the continuation (again averaged over all tokens) to handle more open-ended generation cases. This can happen in zero-shot QA because without any formatting constraint, the

model can output a reasoning path without giving an explicit final answer. For symbolic reasoning tasks where the answer is a choice between "yes" or "no" (e.g., Coin Flip, Web of Lies), we compute the difference between the probabilities masses over "yes/true" and "no/false" (cases ignored). We found when the answer choices are fixed, processing the continuation in this way is slightly more accurate than computing Δ over the continuation directly, since sometimes the model might output invalid options like "We don't know" with high confidence. Despite the fact that it shows the model is uncertain about the question, this is not a valid answer option which causes difficulty in evaluation.

Remove ill-formed responses. Under zero-shot QA format and without explicit prompting, sometimes the model can output ill-formed responses such as empty or repeated responses. Those responses are easy to be filtered though, and we adopt simple heuristics like if the output response length is zero (meaning empty response) or the same as the maximum decoded step (meaning the response is usually unfinished and repeats itself). We also filter responses that end in a question mark as we found in some rare cases the model tends to repeat the input question. For Mistral models we found in some cases the model outputs texts similar to the training data in alternative decoded paths (similar to the findings in Nasr et al. (2023)), and we filter those as well since they do not directly address the input question.

Experiment settings for the Mistral model. For the Mistral pre-trained model, we format the question similarly as "Q: question\nA:". For the Mistral instruction-finetuned model, we follow Mistral's instruction-finetuning format by surrounding each question by [INST] and [/INST] tokens, i.e., "[INST] question [/INST]".[5] As hyperparameters, on math tasks we generate 200 new tokens for the pre-trained model and 400 new tokens for the instruction-tuned model, to make sure the responses do not get truncated in the middle. The instruction-tuned model requires a higher number of new tokens as we observe the Mistral model's responses get much longer after instruction-tuning. For the year parity task, we generate 50 new tokens for the pre-trained model and 100 new tokens for the instruction-tuned model.

Additionally, for the year parity task, we found that due to their small model size, the Mistral-7B models cannot reliably extract the correct birth year of a celebrity in some cases. Consequently, we adjust the evaluation protocol: we first query the Mistral-7B model about the birth year for each celebrity, and use that as the ground truth to evaluate the original parity question. Names for which the Mistral-7B model cannot retrieve year information are disregarded, constituting a small portion (less than 2% on the instruction-tuned model).

Compute Resources. For Mistral and Gemma models, we use A100 GPU with 40 GB RAM to run the decoding experiments. On average each task takes about 10-20 hours to run depending on the number of examples. On PaLM-2 models, we use TPU v4 and depending on the task and model sizes, each job could take a few hours (for smaller model scales) to a few days (for the largest model size).

[5] https://huggingface.co/mistralai/Mistral-7B-Instruct-v0.1